CUSTOMER SALES AND REVENUE ANALYSIS

Using Online Retail II Dataset

PROJECT REPORT

Submitted by

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Data Analytics Project

Abstract

This project analyses retail transaction data from the Online Retail II dataset using Python. The work includes inspecting the dataset, checking data quality, preparing transaction records for sales analysis, calculating revenue, and exploring sales patterns by time, product, country, and customer. Pandas is used for data handling and cleaning, while Matplotlib is used to create visualisations. The dataset loaded for this project contains 1,067,371 rows and 8 columns, with invoice dates ranging from 1 December 2009 to 9 December 2011. Final numerical findings should be inserted from the completed notebook outputs before submission.

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1. Introduction

Data analytics is the process of examining data to discover useful information and support decision-making. Retail businesses can use transaction data to understand product demand, customer purchasing behaviour, revenue patterns, and sales performance.

This project uses the Online Retail II dataset and Python to inspect, clean, and analyse retail transactions. The analysis focuses on data quality and descriptive sales analysis, including monthly revenue, top-performing products, country-wise revenue, and customer purchasing patterns.

2. Problem Statement

Retail transaction datasets can contain missing values, duplicate rows, cancelled invoices, returns, and invalid quantity or price values. If these issues are not considered, sales summaries may be misleading. This project examines the quality of the available records and prepares a suitable dataset for sales analysis.

3. Objectives

- Understand the structure and contents of the Online Retail II dataset.
- Inspect missing values and duplicate records.
- Identify cancelled invoices and transactions with non-positive quantities or prices.

- Prepare a cleaned dataset for descriptive sales analysis.
- Calculate revenue using quantity multiplied by unit price.
- Explore monthly sales, product performance, country-wise revenue, and customer activity.
- Present findings through charts and provide practical business recommendations.

4. Dataset Description

The project uses the Online Retail II Excel workbook. In the notebook, the sheets named “Year 2009-2010” and “Year 2010-2011” were combined for analysis.

Dataset detail	Value
Rows loaded	1,067,371
Columns	8
Date range shown in notebook	1 December 2009 – 9 December 2011
Invoice	Invoice number
StockCode	Product or stock code
Description	Product description
Quantity	Quantity purchased
InvoiceDate	Date and time of transaction
Price	Unit price
Customer ID	Customer identifier
Country	Customer country

5. Tools and Technologies

- Python: programming language used for analysis.
- Google Colab: cloud-based notebook environment used to run Python code.
- Pandas: used to load, inspect, clean, group, and summarise data.
- Matplotlib: used to create charts and visualise patterns.
- Excel workbook: source file format for the dataset.

6. Methodology

1. **Data loading:** Read the two Excel sheets and combined them into one DataFrame.
2. **Data inspection:** Checked the number of rows and columns, column names, sample records, data types, and date range.
3. **Data quality assessment:** Checked missing values, duplicate records, cancelled invoices, and non-positive quantities or prices.
4. **Data preparation:** Created a working copy, removed duplicate rows, handled missing product descriptions where needed, and prepared transaction data for sales analysis.
5. **Revenue calculation:** Calculated revenue as Quantity × Price.

6. **Exploratory analysis:** Grouped data to examine monthly revenue, product performance, country-wise revenue, and customer activity.
7. **Visualisation and export:** Created charts and exported the cleaned sales dataset to CSV.

7. Data Quality Checks and Cleaning

The data quality review considered missing product descriptions, missing customer IDs, duplicate rows, cancelled invoices, and transactions with zero or negative quantities or prices. These checks help identify records that may not be suitable for ordinary sales calculations.

For the sales analysis, a separate sales dataset was prepared by excluding invoice numbers beginning with “C” and keeping records with Quantity greater than zero and Price greater than zero. Revenue was then recalculated as Quantity multiplied by Price. This is a sales-analysis filter; it should not be interpreted as a complete audit of all returns or cancellation reasons.

The original dataset should be preserved, and all cleaning rules should be documented so the analysis can be reproduced. Missing customer IDs were not automatically treated as errors because those rows may still contain useful product and revenue information.

8. Exploratory Data Analysis

8.1 Overall sales summary

Summarise total revenue, number of invoices, number of identified customers, and total quantity using the cleaned sales dataset.

8.2 Monthly sales trend

Group revenue by month and use a line chart to observe changes over time. Identify the month with the highest revenue from the notebook output.

8.3 Product performance

Rank products by revenue and by quantity sold. Compare the highest-performing products.

8.4 Country-wise revenue

Group revenue by country and chart the leading countries. Interpret results in the context of the dataset.

8.5 Customer analysis

Examine revenue by customer and, if calculated, identify repeat purchasing patterns.

9. Key Findings

Complete this section using the exact outputs from the final Google Colab notebook. Do not estimate or invent values. Replace the bracketed fields below with the results produced by your analysis.

- Total sales revenue: [insert value and currency unit].
- Number of invoices in the filtered sales dataset: [insert value].
- Top product by revenue: [insert product and revenue].
- Top country by revenue: [insert country and revenue].

- Highest-revenue month: [insert month and revenue].
- Customer insight: [insert a result supported by the customer analysis].
- Data quality issues identified: [insert the missing-value and duplicate counts from the notebook].

10. Recommendations

- Monitor monthly sales trends to help plan stock levels and promotions.
- Review high-revenue and high-volume products when making inventory decisions.
- Compare country-wise performance before deciding where to focus marketing or sales efforts.
- Improve the completeness of customer IDs where possible to support customer-level analysis.
- Maintain documented checks for duplicates, cancellations, returns, and invalid quantity or price values.
- Repeat data quality checks whenever new transaction data is added.

11. Limitations and Future Scope

The analysis is descriptive and depends on the accuracy and completeness of the source data. Some transactions do not have customer IDs, and excluding cancellations or non-positive quantities may change the totals compared with the original transaction ledger. Revenue calculated as Quantity × Price is a transaction value measure and should not automatically be interpreted as profit.

Future work could include a more detailed return analysis, customer segmentation, cohort analysis, seasonality analysis, an interactive dashboard, and forecasting of future sales.

12. Conclusion

This project demonstrates a basic data analytics workflow using the Online Retail II dataset. The work combines data inspection, data quality checks, cleaning, revenue calculation, exploratory analysis, and visualisation using Python, Pandas, and Matplotlib. The resulting analysis can help describe retail sales patterns and support data-informed decisions. The final report should include the exact numerical findings and charts from the completed notebook before submission.

Appendix: Code and Visualisations

Insert screenshots or exported images from the notebook here. Recommended figures are: monthly revenue trend, top products by revenue, revenue by country, and product quantity comparison. Add a short caption under each chart explaining what it shows.

Also include the final cleaned CSV filename (cleaned_retail_sales.csv) if it is part of the submission, and add the dataset source/citation required by your course or internship guidelines.